EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)
Homework 6 – Downscaling PRD Temperature in the Extended Typhoon Season
Total Score: 60

Learning Objectives:

- Train a Convolutional Neural Network to downscale ERA5 2 m temperature from 0.25° to 0.1°, matching ERA5-Land.
- Identify typhoon-affected days using meteorological criteria and examine conditional model performance.
- Perform sensitivity analysis by perturbing typhoon-relevant inputs and measuring the temperature response.
- Characterize outlier predictions and assess model robustness across the extended typhoon season.

A downscaled meteorological field allows finer structures to be revealed within each cell, as the cells average over a narrower range of elevation and land type. Classical downscaling approaches include dynamical downscaling, which runs a nested regional climate model at high resolution and is physically complete but computationally expensive, and statistical downscaling, which fits regression or analogue relationships between coarse and fine fields and is cheap but limited to linear or hand-crafted mappings.

In this assignment, you will construct a convolutional neural network that performs spatial-to-spatial downscaling: the model ingests coarse synoptic fields together with static terrain descriptors over the Pearl River Delta and outputs a full 2D daily-mean 2 m temperature map for the extended typhoon season, May to September. Although ERA5-Land already provides 2 m temperature at 0.1°, the exercise targets the mapping rather than the product. The same approach may also transfer to CMIP6 output for future emission scenarios, to other regions, to other meteorological variables, and to even finer target resolutions. One key advantage of applying machine learning to downscaling is its efficiency, running within milliseconds per day, against minutes to hours for nested regional model, an advantage that matters for ensembles and sensitivity sweeps.

Part 0. Dataset Specifications

Dataset	Variable(s)	Spatial Resolution	Temporal Coverage
ERA5 Reanalysis	7 surface/column fields	0.25° × 0.25°, ~25 km	2016-2025, May–Sep, hourly
ERA5-Land	2 m temperature (land-only)	0.1° × 0.1°, ~11 km	2016-2025, May-Sep, daily mean
SRTM v3 (2013)	Elevation (static)	1 arc-second, ~30 m	Static

References:

Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730), 1999–2049. https://doi.org/10.1002/qj.3803. (Data retrieved from: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=analysis_ready_data.)

Muñoz-Sabater, J., et al. (2021). ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. Earth Syst. Sci. Data, 13, 4349–4383. https://doi.org/10.5194/essd-13-4349-2021. (Data retrieved from: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land)

NASA JPL (2013). NASA Shuttle Radar Topography Mission Global 1 arc second [Data set]. NASA EOSDIS Land Processes DAAC. https://doi.org/10.5067/MEaSUREs/SRTM/SRTMGL1.003 (Distributed by OpenTopography: https://doi.org/10.5069/G9445JDF; accessed 2026-10-20)

We have processed the dataset for you. All arrays are stored in

`/lustre/project/eesc4550/shared/hw6/data/hw6_downscaling_data_t2m.npz`

Input channels `X_train, X_val` (110-116°E, 20-25°N)

ERA5 Reanalysis, B × 10 × 21 × 25 at 0.25°

Ch	Name	Variable	Unit	Physical role
0	`msl_day`	Mean sea-level pressure	hPa	Synoptic regime; typhoon flag
1	`tcwv_day`	Total column water vapour	mm	Column moisture; longwave warming
2	`u10_day`	10 m zonal wind	$\mathrm{m\ s^{-1}}$	Zonal advection, sea breeze
3	`v10_day`	10 m meridional wind	$\mathrm{m\ s^{-1}}$	Meridional advection, cold surges
4	`tcc_day`	Total cloud cover	$0-1$	Diurnal range control
5	`d2m_day`	2 m dewpoint temperature	K	Near-surface humidity; Tmin
6	`t2m_w0`	2 m temperature, 00-06 UTC	K	Morning warming
7	`t2m_w1`	2 m temperature, 06-12 UTC	K	Afternoon peak
8	`t2m_w2`	2 m temperature, 12-18 UTC	K	Evening into early night cooling
9	`t2m_w3`	2 m temperature, 18-24 UTC	K	Pre-dawn minimum

Static Terrain `geo_features`, 7 × 100 × 120 at 0.05°

Ch	Name	Variable	Unit	Physical Role
0	`elevation`	SRTM elevation	$\mathrm{km}\ (\div 1000)$	Lapse-rate cooling; 0 over ocean
1	`slope`	Terrain gradient	$\div 100$	Forced ascent, drainage flows; 0 over ocean
2	`coast_distance`	Euclidean distance to coast	$0-1$	Sea-breeze and maritime influence
3	`lat_norm`	Latitude, normalized	$-1\ldots1$	Positional prior
4	`lon_norm`	Longitude, normalized	$-1\ldots1$	Positional prior
5	`land_fraction`	Land–ocean mask	$0\ /\ 1$	Distinguishes land from sea cells
6	`sub_grid_residual`	Fine minus coarse elevation	$\div 100$	Ridge/valley relief invisible to ERA5; 0 over ocean

Target `y_train, y_val`, B × 50 × 60 at 0.1°, (110.1-116.0°E, 20.0-24.9°N)

Name	Variable	Unit
`t2m`	ERA5-Land daily-mean 2 m temperature	°C

Masks (`1 = land`, `0 = ocean`)

Key	Grid	Definition	Purpose
`y_land_mask`	50 × 60	ERA5-Land	`NaN` over ocean as ERA5-Land is a land-only product.
`srtm_land_mask`	100 × 120	SRTM raw elevation > 0.5 m	`NaN` over ocean as radar fails to retrieve elevation over ocean.
`loss_mask`	50 × 60	`y_land_mask` ∧ `upsample(srtm_land_mask)`	***Apply before evaluation to exclude samples that involves unknown values in ERA5-Land and SRTM***. Restricts the evaluation metric to the cells that can test the model's ability.

Part 1. Train and Baseline Comparison

Question 1 (12 pts). Train a 3-layer simple CNN that takes the 10-channel coarse atmospheric field:

- First convolutional layer: 16 filters of size 3×3 on the 10 input channels, stride 1, padding 1 (spatial size preserved at 21×25), followed by `ReLU`.
- Second convolutional layer: 16 filters of size 3×3 on the 16 feature channels, stride 1, padding 1 (still 21×25), followed by `ReLU`.
- Resize the 16 × 21 × 25 feature map to 16 × 50 × 60 by bilinear upsampling without learned parameters.
- Third convolutional layer: 16 filters of size 3×3 on the 16 upsampled channels, stride 1, padding 1, followed by `ReLU`.
- Output convolutional layer: a single 1×1 convolution collapsing 16 channels to 1, producing the final 50 × 60 temperature map.
- Optimizer: `AdamW`, cosine learning-rate schedule, gradient clipping at 1.0, 80 epochs. Use MSE loss on `loss_mask`.

How many trainable parameters are there in the model? Show all the steps.
Report RMSE, MAE, and R^2 on `loss_mask` for the validation set, using the model at epoch with the best validation RMSE.

Question 2 (5 pts). Construct a bilinear-interpolation baseline. For each fine-grid cell, predict a bilinearly upsampled version of the coarse ERA5 daily-mean 2 m temperature, corrected by a per-cell training-set bias:

i. Upsample `X_train[:, 6:10].mean(1) - 273.15` from 21×25 to 50×60 using bilinear interpolation, which gives `bilin_train[i,j]`.
ii. Compute the per-cell bias `bias[i,j]`, which is calculated by taking the mean over training days of `(y_train[i,j] - bilin_train[i,j])`, on `loss_mask` cells only.
iii. Upsample `X_val[:, 6:10].mean(1) - 273.15` the same way, add the bias map to obtain a downscaled prediction over the validation days.

Report RMSE, MAE, R^2 on `loss_mask` with the bilinear-interpolation baseline. Report the relative improvement: `(RMSE_bilinear - RMSE_CNN) / RMSE_bilinear × 100%`. Is the CNN meaningfully better than interpolation?

Part 2. Evaluation of Conditional Skill

Question 3 (8 pts). Define typhoon-affected days in the validation set using meteorological criteria:

- Mean sea-level pressure $\mathrm{msl_day} < 1000\ \mathrm{hPa}$
- Wind speed $\sqrt{u_{10}^2 + v_{10}^2} > 5\ \mathrm{m\ s^{-1}}$
- Total column water vapour $\mathrm{tcwv_day} > 55\ \mathrm{mm}$

Report how many typhoon days you identify in the validation set. Now, split the validation set into typhoon vs non-typhoon days. Report RMSE for each group.

Question 4 (5 pts). Break the validation set down by month (May, Jun, Jul, Aug, Sep). Report RMSE and the prediction bias (observation subtract prediction) for each month. Produce a bar chart on the prediction bias.

Part 3. Sensitivity Analysis

Question 5 (12 pts). Sort the days in the validation set by daily-mean domain-averaged MSL. Select the 2 lowest and the 2 highest MSL days for the sensitivity testing.

To conduct the sensitivity analysis, we consider for each day, and for each of the four channels (`msl_day`, `tcwv_day`, `u10_day`, `v10_day`) in the day, perturb that channel only by $+1\sigma$ and then -1σ in normalized space, holding the other three at their observed values.

Moreover, consider the "typhoon-like" perturbation: `msl` -1σ, `tcwv` $+1\sigma$, `wind speed` $+1\sigma$.

Run the 40 forward passes and record (i) the unperturbed prediction and (ii) the 2x4+1=9 perturbed prediction for each day. Record ΔT = mean over `loss_mask` of `(T_perturbed - T_cnn)` in °C, which is the change induced by a perturbation, and thus decide whether the 2 typhoon-days (with low MSL) have a stronger sensitivity than the 2 non-typhoon days. Also, generate a plot with 4 days x 9 predictions, which is the downscaled map under different scenarios, with ΔT on the legends of each smaller plot. Make sure you label the plot clearly with the date and the perturbation setup.

Part 4. Improving the Baseline

Question 6 (8 pts). The coarse ERA5 temperature should roughly agree with the fine ERA5-Land temperature when both are averaged to the coarse grid, after removing the climatological product bias. Add the physical constraint

$$\mathcal{L}_{\mathrm{cc}} = \mathrm{MSE}\left(\overline{T_{\mathrm{fine}}}^{\mathrm{pooled}} - \overline{T_{\mathrm{fine}}}^{\mathrm{train}} - \left(\overline{T_{\mathrm{coarse}}} - \overline{T_{\mathrm{coarse}}}^{\mathrm{train}}\right)\right)$$

and train two models with $\lambda_{cc} = 0.1$ and 1.0, respectively. Here, the MSE is the mean-squared error calculated by taking the average over the number of samples in a batch.

$\overline{T_{\mathrm{fine}}}^{\mathrm{pooled}}$ represents the current batch's prediction, averaged to 21×25 with

`F.adaptive_avg_pool2d(pred, (21, 25))`;

$\overline{T_{\mathrm{fine}}}^{\mathrm{train}}$ represents the training-set climatology (time-average over the 1224 training days) of the fine target, pooled the same way as $\overline{T_{\mathrm{fine}}}^{\mathrm{pooled}}$;

$\overline{T_{\mathrm{coarse}}}$ represents current batch's daily-mean ERA5 t2m in °C;

$\overline{T_{\mathrm{coarse}}}^{\mathrm{train}}$ is the training-set climatology of the coarse field.

Report RMSE, MAE, and R^2 on `loss_mask` for the validation set, using the model at epoch with the best validation RMSE for the two physics-informed models.

Question 7 (10 pts). Improve the simple CNN architecture to achieve a validation RMSE of 0.3°C. You may also try to train 3 models with different random seeds and average their predictions at inference. It is known that such a method can average out the prediction error.
Report your final model description, validation set metrics (RMSE, MAE, R^2).
Note: You will receive **partial scores** even you cannot reach the prescribed target. Any change to the dataset, split, target definition, or methods that cause data leakage will result in **zero** points for this question, regardless of the reported RMSE.

Submission Guideline: To complete the homework, please submit **TWO** files:

- `hw6_<your_id>.pdf`, which contains the answers to all the **6** questions;
- `hw6_<your_id>.zip`, which contains the Python scripts to produce the included plots and scripts that perform data analysis.

If any AI tool was used in writing or code generation, please include a short note on AI use.

----- **End of Homework 6** -----